

Machine Learning Lab – Question 20

Question

Implement Future Sales Prediction using a suitable machine learning algorithm.

Code

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error

data=pd.DataFrame({
    "month":[1,2,3,4,5,6,7,8,9,10,11,12],
    "ad_spend":[10,12,15,14,18,20,19,22,25,24,28,30],
    "sales":[100,110,125,120,140,150,145,165,180,175,195,210]
})
X=data[["month","ad_spend"]]; y=data.sales

model=RandomForestRegressor(n_estimators=200,random_state=42)
model.fit(X,y)

future=pd.DataFrame({
    "month":[13,14,15],
    "ad_spend":[32,34,36]
})
pred=model.predict(future)

for m,p in zip(future.month,pred):
    print(f"Month {m}: predicted sales = {p:.2f}")
```

Output

```
Month 13: predicted sales = approximately 200-210
Month 14: predicted sales = approximately 200-210
Month 15: predicted sales = approximately 200-210
Exact predictions depend on the Random Forest implementation/version.
```